

Research Article

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Effects of epigallocatechin-3-gallate of *Camellia sinensis* leaves on blood pressure, lipid profile, atherogenic index of plasma and some inflammatory and antioxidant markers in type 2 diabetes mellitus patients: a clinical trial

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Abstract

Background: In patients with type 2 diabetes mellitus (T2DM) the inflammatory and metabolic responses to epigallocatechin-3-gallate (EGCG) are unknown.

Objectives: Evaluate the impacts of EGCG on metabolic factors and some biomarkers of stress oxidative in patients with T2DM.

Methods: In this randomized, double-blind, placebo-controlled trial, 50 patients with T2DM consumed either 2 tablets (300 mg) EGCG (n=25) or wheat flour as placebo (n=25) for 2 months. The total antioxidant capacity (TAC), interleukin-6 (IL-6), lipid profile, mean arterial pressure (MAP), atherogenic index of plasma (AIP) were evaluated before and after the intervention.

Results: The finding of present study exhibited a significant increase in the serum levels of TAC after the EGCG supplementation (p=0.001). Also, in compare with control

group, the mean changes of TAC were significantly higher in supplement group (p=0.01). In intervention group, a significant decrease was observed in the mean levels of triglyceride, total cholesterol, diastolic blood pressure (DBP), AIP, and MAP (p<0.05). Taking EGCG resulted in the mean changes of total cholesterol, MAP and DBP were significantly lower in compare with control group (p<0.05). **Conclusions:** This study recommended that EGCG supplementation may be improved blood pressure, lipid profile, AIP, and oxidative status in patients with T2DM.

Keywords: EGCG; inflammation; lipid profile; type 2 diabetes mellitus.

Introduction

Type 2 diabetes mellitus (T2DM) is a costly chronic disease that is accompanied by an increase in blood sugar levels and insulin resistance [1]. 415 million individuals suffered from diabetes in 2015, with more than 90% having T2DM and will probably reach to 642 million by 2040 [2]. Several clinical features such as hyperglycemia, low-density lipoproteins (LDL), central obesity, prothrombotic status, lipid disorders, oxidation, inflammation, hypertension are associated to T2DM [3]. The role of oxidative stress in the pathogenesis of T2DM is important. An increase in the concentration of carbonyl and sulfidryl proteins, malondialdehyde (important biomarker of lipid peroxidation) and total antioxidant capacity (TAC) reduction have been suggested as markers for oxidative stress [4]. In patients with T2DM, increased production of reactive oxygen species (ROS) and impaired balance between oxidant and antioxidant enzymes causes a pro-inflammatory condition. In fact, the generation of various pro-inflammatory markers increases insulin resistance and hyperlipidemia

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in diabetic patients [5]. Due to the importance of oxidative stress in progression and the development of T2DM and its vascular complications, the consumption of natural compounds with antioxidant properties as a preventive and therapeutic method for this disease is very interesting [6]. Treatment with green tea is one of these therapeutic approaches. Catechins are the main components of tea polyphenols [7]. Epigallocatechin-3-gallate (EGCG) is main catechin in green tea (50–80%) that most of the antioxidant and anti-inflammatory properties related to consumption of green tea have been linked to it [8]. EGCG by inhibiting the production of the microsomal enzyme 11 β -hydroxysteroid dehydrogenase type 1 (11 β -HSD1) (cortisol hormone activator) declines local cortisol concentrations and improves blood pressure, lipid profile and glycemic control [9]. Moreover, EGCG through appetite suppression, body weight loss, and release of endocrine pancreatic hormones improves glucose tolerance and enhances insulin sensitivity [10]. EGCG acts as an antioxidant and anti-inflammatory agent in various pathological disorders including diabetes and kidney disease [11]. Different evidence indicates that EGCG could promote the activity of antioxidant enzymes such as glutathione peroxidase (GPx), and superoxide dismutase, inhibits pro-oxidant enzymes and protects cells from ROS-induced damages [12]. According to the above effects, the hypothesis of present study was that the use of EGCG is effective in improving the condition of diabetic patients against the lack of effect. To our best of knowledge, there are few studies that have evaluated the effects of EGCG on patients with T2DM. So, because of inadequate evidence present study was done to investigate that consumption of green tea EGCG improves the lipid profile, atherogenic index of plasma (AIP) and some inflammatory and antioxidant markers in patients with T2DM.

Methods

Study design and subjects

In the clinical trial, 100 patients with T2DM were recruited from outpatient endocrinology and metabolism clinics of Golestan referral and governmental Hospital of Ahvaz Jundishapur University of Medical science in Ahvaz, Khozestan, Iran, during August 2017 to March 2018 (Figure 1). 50 patients were excluded from study because of declined to participate and not meeting inclusion criteria. Based on inclusion criteria, 50 patients were randomly allocated into the two groups placebo (n=25) and intervention (n=25) (according to random permuted block procedure (block design) based on the combined analysis). The intervention group consumed 2 tablets (300 mg) EGCG of *Camellia sinensis* leaves (Shari Co., Iran) per day for 2 months and the control group consumed 2 placebo tablets (300 mg) containing wheat flour (made by the Faculty of Pharmacy in Ahvaz Jundishapur University of

Medical Sciences). Inclusion criteria were as follows: BMI between 18.5 and 35 kg/m², aged 20–60 years old, and diabetes duration of more than 2 years. Exclusion criteria including: pregnancy and lactation, kidney failure, digestive disorder, liver damage, cardiovascular and inflammatory diseases, thyroid disease, cancer, the use of antioxidant supplements during the last 6 months, insulin injection.

Anthropometric, blood pressure and nutritional assessments

The body weight (accuracy of 0.1 kg) and height (accuracy of 0.1 cm) were evaluated by a standard scale (Seca). Formula: (weight (kg))/(height² (M)) was used for calculation of BMI. After 20 min of rest and sitting on the chair, blood pressure was measured. The formula: (2DBP + SBP)/3 was used for calculation of MAP. To assessment the dietary intakes of patients, a 3-Day food records questionnaire was used. Nutritionist 4 was used for analysis of dietary intake.

Biochemical evaluation

5 mL blood sample were collected in the fasting condition pre/post intervention. The levels of lipid profile were measured immediately after the sampling and the remnants of serum were stored at –80 °C for later assessment. The lipid profiles were determined by auto-analyzer. The AIP was calculated with the logarithm TG/HDL-c [13]. Enzyme-linked immunosorbent assay (ELISA) method was used for measurement of serum IL-6 (Diaclone, Beaucou, France) and TAC (Zell Bio GmbH kit, Germany).

Statistical analysis

All analyses were done with SPSS version 20.0. A p-value<0.05 was considered as significant. Data were stated as mean $\pm$ SD. The Kolmogorov–Smirnov test was used to evaluate the normality of the data. In order to evaluate within groups differences (before and after the intervention), paired sample t-test was used. The Independent t-test was used to compare qualitative data between the two groups before and after the intervention. Qualitative data were compared with Chi square test. The sample size was calculated with 90% power test at a significance level of 5%, with a 95% confidence interval and considering TAC as the main variable in the study by Toolsee et al. [14], a sample size of 16 patients was calculated in each group using the following formula:

$$n = \frac{(Z_{1-\alpha/2} + Z_{1-\beta})^2 (S_1^2 + S_2^2)}{(\mu_1 - \mu_2)^2}$$

Sample size was increased to 24 participants in each group considering possible withdraw.

Results

In the present study, all results had a normal distribution. Forty-four patients (n=22 in intervention group, n=22 in control group) were finally analyzed for this study. In terms

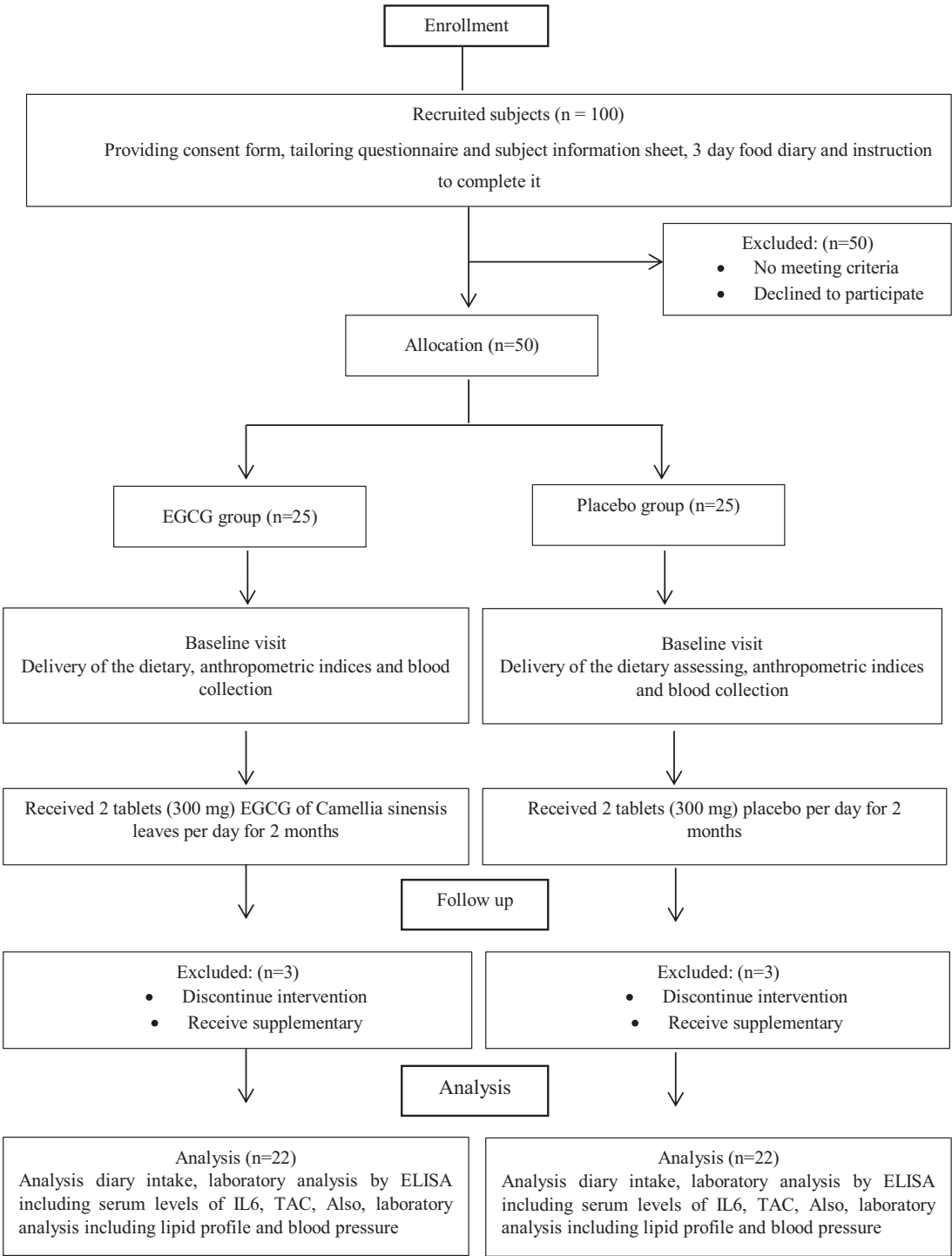


Figure 1: Flow chart of the study.

of age, there was no significant difference between the two groups of intervention and control (51.75 ± 6.79 and 52.61 ± 7.22 years, respectively). At the beginning of the study, no difference was observed ($p \geq 0.05$) in gender, BMI, education level and physical activity between the two groups of intervention and control (Table 1).

At the beginning of the study, the results did not show significant difference ($p \geq 0.05$) in the mean intake of energy, macronutrients (carbohydrates, protein, fat and fiber) and micronutrients between the two groups (Table 2).

There was no significant difference between the two groups in serum levels of IL-6 and TAC at baseline. EGCG

Table 1: Baseline characteristics.

Variables	Intervention group (n=22)	Placebo group (n=22)	p-Value
Age, years	51.75 ± 6.79	52.61 ± 7.22	0.71
Male/Female	11/11	10/12	0.76 ^a
BMI, kg/m ²	29.46 ± 4.28	29.28 ± 3.98	0.89
Diploma or less/University	18/4	20/2	0.38 ^a
Physical activity (met-min/week)	143.31 ± 81.16	132.22 ± 66.37	0.62

All values are means ± SD. Between group comparisons of participants characteristics resulted from Independent sample t-test (for physical activity, age and BMI) and ^aChi-square test (for gender and education). BMI, body mass index.

supplementation significantly increased serum level of TAC (0.33 ± 0.06 vs. 0.44 ± 0.08 mM, respectively; p=0.001). Also, it was seen a significant increase in the mean changes of TAC in the intervention group compared with the control group (0.11 ± 0.08 vs. -0.07 ± 0.31 mM, respectively; p=0.01). The mean serum levels of IL-6 had not significant difference in two control and intervention groups post-intervention (p≥0.05) (Table 3).

According to the analysis between the two groups, no significant differences were observed in serum TG, HDL_c, LDL_c, CHOL, VLDL, and AIP at baseline. EGCG supplementation significantly reduced AIP, TG, CHOL,

Table 2: Dietary intake between two groups.

Variables	EGCG group (n=22)	Placebo group (n=22)	p-Value ^a
Energy, kcal	1862.95 ± 524.78	1906.39 ± 345.73	0.74
Protein, gr	73.23 ± 16.45	75.79 ± 15.27	0.58
Carbohydrate, gr	288.80 ± 91.86	287.94 ± 68.66	0.97
Fat, gr	48.95 ± 25.47	51.19 ± 27.12	0.77
Cholesterol, mg	233.37 ± 291.28	289.27 ± 286.78	0.51
Soluble fiber, mg	0.31 ± 0.22	0.34 ± 0.31	0.68
Zinc, mg	7.03 ± 2.63	8.04 ± 2.28	0.17
Magnesium, mg	223.32 ± 82.12	259.94 ± 118.65	0.23
Manganese, mg	3.36 ± 1.23	3.37 ± 1.21	0.99
Iron, mg	16.29 ± 7.19	14.66 ± 4.59	0.36
Sodium, mg	836.44 ± 595.11	1265.11 ± 1013.22	0.09
Vitamin A, mcg	522.11 ± 440.97	653.55 ± 583.81	0.39
Vitamin K, mg	109.78 ± 58.69	87.92 ± 77.16	0.28
Vitamin E, mg	2.57 ± 1.93	2.79 ± 1.63	0.69
Vitamin C, mg	83.48 ± 42.72	108.62 ± 48.08	0.07
Vitamin B1, mg	1.95 ± 0.54	1.99 ± 0.47	0.80
Vitamin B3, mg	21.81 ± 6.74	22.32 ± 5.77	0.78
Vitamin B5, mg	4.19 ± 1.40	3.98 ± 1.65	0.65
Vitamin B9, mg	309.17 ± 130.92	344.64 ± 109.60	0.32

All values are means ± SD. ^aBetween group comparisons of participants characteristics resulted from Independent sample t test.

Table 3: Within-group and between-group analysis of EGCG effects.

Variables	EGCG group (n=22)	Placebo (n=22)	p-Value ^a
SBP, mmHg			
Baseline	13.76 ± 2.01	13.25 ± 2.38	0.44
After 2 months	13.38 ± 1.75	13.50 ± 2.39	0.85
p-Value ^b	0.39	0.52	
DBP, mmHg			
Baseline	8.90 ± 1.18	8.14 ± 1.27	0.06
After 2 months	8.47 ± 1.04	8.28 ± 1.25	0.59
p-Value ^b	0.03	0.50	
MAP, mmHg			
Baseline	10.52 ± 1.16	9.84 ± 1.57	0.11
After 2 months	10.11 ± 1.12	10.02 ± 1.47	0.83
p-Value ^b	0.03	0.44	
TG, mg/dL			
Baseline	168.68 ± 87.27	189.13 ± 93.52	0.45
After 2 months	155.22 ± 78.73	193.52 ± 91.13	0.14
p-Value ^b	0.009	0.68	
TC, mg/dL			
Baseline	167.77 ± 30.03	155.31 ± 43.46	0.27
After 2 months	159.59 ± 31.33	158.45 ± 44.47	0.92
p-Value ^b	0.001	0.32	
LDL, mg/dL			
Baseline	83.13 ± 27.69	68.47 ± 26.51	0.08
After 2 months	81.08 ± 30.76	70.73 ± 28.75	0.25
p-Value ^b	0.15	0.36	
HDL, mg/dL			
Baseline	49.27 ± 8.27	49.56 ± 9.88	0.91
After 2 months	49.41 ± 8.40	49.39 ± 10.49	0.99
p-Value ^b	0.84	0.84	
AIP			
Baseline	0.49 ± 0.21	0.54 ± 0.25	0.47
After 2 months	0.45 ± 0.21	0.55 ± 0.25	0.17
p-Value ^b	0.03	0.65	
IL6, pg/mL			
Baseline	2.05 ± 1.15	1.76 ± 1.56	0.48
After 2 months	1.71 ± 0.78	1.83 ± 1.17	0.64
p-Value ^b	0.10	0.78	
TAC, mM			
Baseline	0.33 ± 0.06	0.36 ± 0.14	0.44
After 2 months	0.44 ± 0.08	0.29 ± 0.29	0.02
p-Value ^b	0.001	0.32	

All values are means ± SD. ^aBetween group comparison of variables at baseline and after the intervention resulted from Independent sample t-test. ^bWithin group comparison of variables resulted from Paired sample t-test. BMI, body mass index, SBP, systolic blood pressure; DBP, diastolic blood pressure; MAP, mean arterial pressure; AIP, atherogenic index of plasma; TAC, total antioxidant capacity.

and VLDL in the intervention group, AIP (0.49 ± 0.21 vs. 0.45 ± 0.21, respectively; p=0.03), TG (168.68 ± 87.27 vs. 155.22 ± 78.73 (mg/dL), respectively; p=0.009) and CHOL (167.77 ± 30.03 vs. 159.59 ± 31.33 mg/dL, respectively; p=0.001). Although, serum levels of HDL_c and LDL_c did not change significantly after the intervention.

Table 4: Between-group changes analysis of EGCG effects.

Variables	Δ	Δ	p-Value ^a
	EGCG group	Placebo	
SBP, mmHg	-0.38 ± 2.10	0.24 ± 1.77	0.28
DBP, mmHg	-0.42 ± 0.89	0.14 ± 0.98	0.05
MAP, mmHg	-0.41 ± 0.87	0.17 ± 1.05	0.05
TG, mg/dL	-13.45 ± 21.80	4.39 ± 50.25	0.13
TC, mg/dL	-8.18 ± 9.55	3.13 ± 14.60	0.004
LDL, mg/dL	-2.05 ± 6.49	2.25 ± 11.56	0.13
HDL, mg/dL	0.13 ± 3.15	-0.17 ± 4.20	0.78
AIP	-0.03 ± 0.07	0.01 ± 0.13	0.14
IL6, pg/mL	-0.35 ± 0.96	0.07 ± 1.22	0.20
TAC, mM	0.11 ± 0.08	-0.07 ± 0.31	0.01

All values are means $\pm$ SD. ^aCompare changes between the two groups from Independent sample t-test. SBP, systolic blood pressure; DBP, diastolic blood pressure; MAP, mean arterial pressure; AIP, atherogenic index of plasma; TAC, total antioxidant capacity.

Regarding with CHOL, there was a significant difference in the mean changes between the intervention and control groups after the intervention (-8.18 ± 9.55 and 3.13 ± 14.60 mg/dL, respectively: $p=0.004$) (Table 4).

The results of Table 4 did not show significant differences in SBP, DBP and MAP between the intervention and control groups at the baseline ($p \geq 0.05$). A significant reduction was seen in DBP and MAP after the intervention, DBP (8.90 ± 1.09 vs. 8.47 ± 1.04 mmHg, respectively; $p=0.03$) and MAP (10.52 ± 1.16 vs. 10.11 ± 1.12 mmHg, respectively; $p=0.03$). However, decrease in SBP was not significant in intervention group. In intervention group, compared with the control group, the mean changes of DBP and MAP were lower, and in terms of significant, this differences were marginal, DBP (-0.42 ± 0.89 vs. 0.14 ± 0.98 mmHg, respectively; $p=0.05$) and MAP (-0.41 ± 0.87 vs. 0.17 ± 1.05 mmHg, respectively; $p=0.05$) (Table 4).

Discussion

In present study EGCG supplementation for 2 months significantly improved serum TAC levels. Also, showed a significant increase in the mean TAC changes in the supplement group compared with the control group. A significant decrease was observed in TG, CHOL, AIP, MAP, and DBP in the intervention group. EGCG supplementation significantly reduced the mean changes of CHOL, MAP and DBP compared with the control group.

Previous studies have indicated useful properties of green tea on antioxidant capacity in healthy subjects [15]. Similar to present trial, Basu et al. [16], have suggested that 8 weeks supplementation with green tea in patients with metabolic syndrome improved TAC, other clinical trials

observed similar effect in patients with coronary heart problems [17]. In contrast with our results, supplementation with 560 mg/day green tea extract did not show a significant increase in serum levels of TAC for 12 weeks [18]. Maybe it can be said that one of the reasons for different results is the use of EGCG as major component of green tea in present study. Consumption of EGCG for two months protected the increase in serum levels of IL-6 in diabetic rats, but failed to significantly reduce serum levels IL-6. Also, a remarkable decrease was seen in serum IL-1b, IL-6, and TNF- α in diabetic rats under treatment with EGCG for one month [19]. The difference in target population, the method of study and supplement dose are proposed to be possible reasons leading to the diversity in the outcomes. One of the anti-inflammatory mechanisms of EGCG is that inhabits inflammatory responses through activator of transcription 3 (STAT3) signaling pathways and suppression of NF- κ B (nuclear factor-kappa B) and signal transducer [20]. Moreover, EGCG attenuates the activity of pro-oxidant enzymes and promoted the activity or expression glutathione peroxidase, superoxide dismutase, and catalase as antioxidant enzymes [21].

Dyslipidemia is one of the most important factors for coronary artery disease that develops the progression of atherosclerosis in T2DM. AIP has recently been suggested as a predictive biomarker for plasma atherogenicity and has positive correlation with diabetes, cardiovascular disease risk, and hypertension [22]. The finding of present study exhibited that EGCG supplement reduced the serum levels of TG, CHOL and AIP. It is known that consumption of dietary polyphenols (such as EGCG) as a lipid risk factors lowering strategy play an important role in improvement of heart disease and relevant complications in patients with diabetes [23]. Similar to this study, (2 mg/kg body weight) EGCG supplement in diabetic rats for one month resulted in a noticeable increase in serum HDL levels and significantly decreased triglycerides, total cholesterol, VLDL, and LDL [17]. Also, the studies exhibited that use of green tea leaves extract containing polyphenols antioxidants in hypercholesterolemic animals reduced the atherogenic index (48%) and reduction in this index is considered as an anti-atherosclerotic factor [24]. The results of a meta-analysis demonstrated that green tea (dosage between 39 and 856.8 mg/day) significantly decreased plasma LDL and TC in overweight or obese individuals. However, showed no effect on plasma TG and HDL levels in overweight or obese people [25]. The cholesterol-lowering impact related to EGCG may happen by up-regulation of its receptors in cell livers and inhibition of intestinal cholesterol absorption [26]. Also, regarding the importance of oxidative stress in induction of lipid peroxidation, the existence of

compounds in molecular structure of EGCG such as gallate catechin and hydroxyl groups may be reasons for scavenging activity of EGCG against protein oxidation and ROS-induced lipid peroxidation [27].

Contrary to the finding of this study, a randomized clinical trial on women with syndrome metabolic showed that consumption of green tea drink for 8 weeks significantly decreased SBP, but no significant change was seen in DBP [28]. The difference in the target population and the type of treatment could be the reasons for this dissimilarity. In agreement with our findings, Jiangsu Nutrition Study (JIN) showed that tea intake was inversely correlated with 5 year DBP but not the SBP in smokers, and inversely related to with both SBP and DBP in nonsmokers [29]. Green tea acts by several different possible mechanisms on blood pressure. The studies have demonstrated that EGCG reduces the concentrations of prostaglandin E2 and kallikrein and decreases the actions of NADPH oxidase and angiotensin-converting enzyme (ACE). This action can reduce blood pressure. Generally, it is known that 800 mg/day EGCG had no detectable side effects in subjects [30]. Nevertheless, dehydration is a deleterious effect of overuse tea and its green tea polyphenols (GrTPs). Long-term consumption of GrTP may change the synthesis of bile acid and rise hepatotoxicity through increasing bilirubin levels and alanine aminotransferase [31]. Also, reported that high dose of GrTP consumption (2.6 mg/g) and EGCG (1.3 mg/g) [32] may reduce weight (as beneficial side effect). Failure to check the glycemic status can be one of the limitations of this study.

Conclusion

The results showed that EGCG supplementation may improve lipid profile, blood pressure, biomarkers of stress oxidative in patients with T2DM.

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Informed consent: The informed consent was obtained from all participants.

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